MARIAM ABU-RAHMA

mariammaburahma@gmail.com | 858-790-1297
Portfolio: mariammabu.github.io

EDUCATION

Homestead High School, Sunnyvale, CA | *August 2023-Present*

- **GPA:** 4.0 (unweighted) | 4.51 (weighted)
- **Relevant Coursework:** AP Calculus BC, AP Physics 1, AP Chemistry, AP Computer Science A, AP Computer Science Principles
- **Interests:** Electrical Engineering, Mechanical Engineering, Robotics

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, HTML/CSS
- **Hardware & Electrical:** Arduino microcontrollers, IR communication systems, motor control systems, circuit prototyping
- **Design & Engineering tools:** Altium PCB design, Fusion 360, 3D printing, Github
- **Hobbies:** Needle felting, cooking, 3D modeling and printing, robotics prototyping
    - Art Channel with 850+ subscribers and over 180,000 views
- **Languages**: English, Arabic

PERSONAL PROJECTS

Lead Developer | AIxKumite: Computer Vision for Combat Sports | *June 2025 - Present*
Developing LLM-based systems using YOLOv8 and MediaPipe through coded JSON prompts to analyze karate matches and extract performance data to improve overall player performance. Using computer vision and pose-detection algorithms to identify techniques and evaluate athlete performance.

Hardware Engineer | Arduino IR Control System & PCB Design | *January - June 2026*
Designed and built a handheld infrared device to control LED systems using microcontroller programming. Integrated hardware circuits with software signal processing to transmit commands. Designed custom PCB using Altium emphasizing clear signals and noise reduction for consistent signal processing.

Electrical Engineer | NFC Attendance Tracking System | *April - May 2026*
Designed and built an NFC-based attendance system using an ESP32 microcontroller and PN532 NFC reader to automate student check-ins at my karate dojo. Developed a WiFi-connected system that sends attendance data to Google Sheets through Google Apps Script integration. Improved user experience by adding LED feedback indicators for scan detection, processing status, successful check-ins, and error handling.

RESEARCH EXPERIENCE

Research Intern | Worcester Polytechnic Institute (WPI) | Summer 2026
Worked with PhD students under the supervision of Professor Ulkuhan Guler to develop a wearable sensor system for monitoring oxygen (O_2) and carbon dioxide (CO_2) levels. Designed health monitoring chip through Altium to detect hypoxia and other respiratory conditions

Research Intern | Basys.ai (Healthcare AI) *June - August 2026*
Worked with a team of 10 interns researching artificial intelligence systems used in healthcare. Studied how LLMs analyze patient electronic health records (EHR) and examined risks such as hallucinated medical outputs focusing on data training with poor quality and LLM's inconsistency in outputs. Collaborated with a group of 5-10 interns through daily meetings and technical discussions.

Robotics Program | MIT BeaverWorks (BWSI) *October - December 2025*
Programmed an autonomous vehicle using computer vision and line-following algorithms. Used OpenCV and HSV color detection to identify track markers and guide the car. Developed algorithms to adjust steering and speed based on detected errors. Team project earned 2nd place out of 10 teams and Academic Achievement recognition.

Volunteer English Tutor | Bay Area Tutoring Association | June-August 2024
Tutored Spanish speaking elementary school students English skills. Worked one-on-one and used IXL to teach reading, vocabulary, grammar, and writing.

Lead Researcher | Journal of Emerging Investigators (JEI) *January - April 2024*
Co-Designed and conducted a cross-sectional survey study on hygiene behaviors after COVID-19 among high school students in Punjab, Pakistan and Santa Clara County, California (N=174) using Google Forms and social media outreach. Analyzed responses using two-proportion Z-tests in Excel to compare trends across gender and regions. Found that 28–36% of students no longer wore masks and 40–56% had returned to their pre-pandemic hygiene habits, showing that many students had stopped following COVID guidelines over time. Co-authored a 15-page manuscript and published it through the JEI peer-review process.

LEADERSHIP & TEACHING

Muslim Student Association (MSA) Vice President | Homestead HS *2025 - Present*
Assist in coordination of events and activities such as budgeting, hosting speakers, and holding kahoot activities that support community engagement and leadership.

Founder & Curriculum Developer, STEM Outreach Initiative | Cairo, Egypt *July 2025*
Ran an independent engineering outreach program, delivering workshops in Arabic to students in under-resourced communities. Curriculum featured motor-powered vehicles and LED circuitry to introduce fundamental Electrical and Mechanical Engineering concepts. Managed all logistics and instruction for 90-minute sessions, adapting complex engineering principles for 10 students ages 6–12 to bridge gaps in STEM education.

Karate Instructor & Mentor | Sunnyvale Martial Arts *January - April 2025*
Coached 20 younger athletes and assisted with training sessions, helping students improve sparring techniques using strategies I have learned from competing while exhibiting confidence during tournaments.

HONORS & INTERNATIONAL RECOGNITION

World Karate Federation (WKF) | Global Rank: #2

- **4x USA National Team Athlete:** Represent Team USA in elite international events
- **Gold Medalist:** 2025 Karate 1 (K1) Series – Monterrey
- **Silver Medalist:** 2024 Karate 1 (K1) Series – Cancun
- **Bronze Medalist:** 2024 Pan American Championships – Brazil

Participant | ISSCC Women's Networking Initiative *February 2026*
Connected with professors from Berkeley, MIT, WPI, BU +, college students, and industry leaders in electrical engineering, and attended presentations regarding Integrated Circuit (IC) design.

University of California Nominee in ELC Program | Homestead High School *2026*
Selected by University of California and Homestead High School for Eligibility in the Local Context (ELC), recognizing top academic performance within the graduating class and eligibility consideration for the University of California system.

7x Green & White Award Recipient | Homestead High School *2024 - 2026*
Recognized annually for demonstrating excellence and well roundedness in class settings while balancing international athletic commitments through math, physics, chemistry, engineering, social studies, and sports classes.